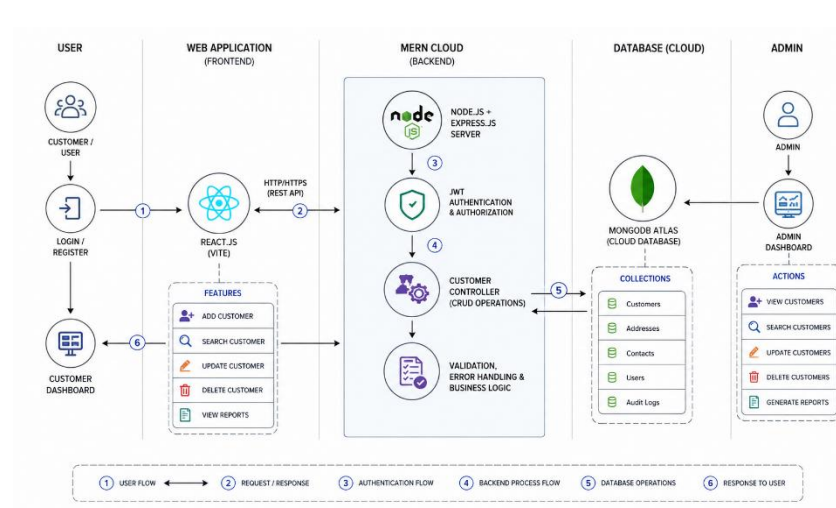


Project Design Phase-II Technology Stack (Architecture & Stack)

Date	13 July 206
Team ID	SWTID-2026-9658
Project Name	Customer Registry
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Single Page Application (SPA) dashboard for ticket submission, tracking logs, and administrative filtering.	React.js, Tailwind CSS, JavaScript
2.	Application Logic-1	Web application routing, server-side data sanitization, middleware logic, and request handling	Node.js, Express.js framework.
3.	Application Logic-2	Object-Document Mapping (ODM) logic to define data models, strict schemas, and perform operations.	Mongoose library.
4.	Application Logic-3	Token tokenization and secure middleware validation for session safety.	jsonwebtoken (JWT) library.
5.	Database	Local non-relational document data storage engine for rapid development environments	MongoDB Community Server
6.	Cloud Database	Cloud-hosted, distributed database instance for live application persistence.	MongoDB Atlas Cluster.
7.	File Storage	File upload storage requirements for image evidence attachments on tickets.	Cloudinary API / Multer local storage engine.
8.	External API-1	Automated dispatch system for email registration validation and ticket confirmations.	Nodemailer service API.
9.	External API-2	<i>Not required for initial MVP version / Left blank.</i>	N/A
10.	Machine Learning Model	<i>Not required for initial MVP version / Left blank.</i>	N/A

11.	Infrastructure (Server / Cloud)	Local development execution and remote multi-region cloud deployment.	Local: Node runtime environment. Cloud: Render platform hosting / Vercel.
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Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	End-to-end framework stack components utilized to ensure zero software licensing acquisition fees.	React, Node.js, Express, Mongoose.
2.	Security Implementations	Role-based authorization layers, validation checks, and secure hashed password storage.	JSON Web Tokens (JWT), Bcrypt password hashing.
3.	Scalable Architecture	Clean decoupling of user dashboard interfaces from server-side query processing steps.	3-Tier Distributed Architecture (React Frontend → Node Backend → MongoDB Database).
4.	Availability	Maintaining persistent connection sessions and zero single points of backend failure.	MongoDB Atlas multi-region replication clusters.
5.	Performance	Low latency client updates, state adjustments, and index-based quick lookups.	React state management re-rendering and MongoDB database indexing.